

AFONSO RAFAEL MARTINS

Senior Machine Learning Engineer

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PROFESSIONAL SUMMARY

Experienced Machine Learning Engineer with 12 years of expertise in designing, developing, and deploying production-grade ML systems at scale. Proven track record of leading cross-functional teams and delivering innovative solutions that drive business value. Specialized in deep learning, NLP, and computer vision with strong proficiency in cloud infrastructure and MLOps practices. Passionate about mentoring junior engineers and establishing best practices across ML organizations.

WORK EXPERIENCE

Staff Machine Learning Engineer

Unbabel

Lisbon, Portugal

www.unbabel.com

January 2021 - Present

Led the design and implementation of a transformer-based NLP model for real-time translation quality estimation, improving accuracy by 23% and reducing inference latency by 40%. Managed a team of 5 ML engineers, overseeing model development, experimentation frameworks, and deployment pipelines across multiple production environments. Architected a distributed training infrastructure using Kubernetes and Ray, enabling 3x faster model training cycles and supporting models with 7B+ parameters. Established ML best practices documentation, code review standards, and created comprehensive onboarding materials that reduced new engineer ramp-up time from 8 weeks to 4 weeks.

Senior Machine Learning Engineer

Synapse Data Ltd

Porto, Portugal

www.synapsedata.io

March 2020 - December 2020

Developed and optimized machine learning models for customer churn prediction, achieving 92% AUC and enabling proactive retention campaigns that recovered 15M EUR in annual revenue. Implemented a feature store using Delta Lake and Spark, standardizing feature engineering across 8 different teams and reducing feature development time by 50%. Conducted extensive research on graph neural networks for recommendation systems, publishing findings internally and presenting at 2 industry conferences. Mentored 3 junior data scientists in production ML practices, code optimization, and model evaluation methodologies.

Machine Learning Engineer

Critical Manufacturing

Covilha, Portugal

www.criticalmanufacturing.com

September 2017 - February 2020

Built and deployed computer vision models for defect detection on manufacturing production lines, achieving 97% precision and reducing manual inspection labor costs by 35%. Designed and implemented an automated retraining

pipeline that monitored model drift, triggered retraining workflows, and deployed updated models to edge devices with zero downtime. Created comprehensive ML monitoring dashboards using Grafana and Prometheus, tracking model performance metrics, data drift indicators, and system health across 12 production environments. Led the migration of legacy batch prediction systems to real-time inference architecture, reducing prediction latency from 2 hours to under 100 milliseconds.

EDUCATION

Master of Science in Computer Science

Universidade de Lisboa, Instituto Superior Tecnico

Lisbon, Portugal

Graduated: June 2012

Specialization: Machine Learning and Artificial Intelligence

Bachelor of Science in Computer Science

Universidade de Lisboa, Instituto Superior Tecnico

Lisbon, Portugal

Graduated: June 2010

TECHNICAL SKILLS

Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers

Programming Languages: Python, SQL, Java, Scala, C++

Deep Learning: Convolutional Neural Networks, Recurrent Neural Networks, Transformers, Graph Neural Networks, Diffusion Models

NLP and Computer Vision: BERT, GPT models, YOLO, ResNet, semantic segmentation, object detection, text classification

Cloud Platforms and Infrastructure: AWS (SageMaker, EC2, S3), Google Cloud Platform (Vertex AI, BigQuery), Docker, Kubernetes

MLOps and Deployment: MLflow, Kubeflow, Feature stores, Model monitoring, A/B testing frameworks, CI/CD pipelines

Data Processing: Apache Spark, Pandas, Dask, Kafka, Apache Airflow

Data Visualization and Analysis: Matplotlib, Seaborn, Plotly, Jupyter Notebooks, Tableau

LANGUAGES

Portuguese: Native

English: Fluent

Spanish: Conversational